the trading strategy). Here the tracking error is identical to the standard deviation of the regression equation:

$$Index\ Adj\ Cost = Arrival\ Cost - \hat{b}_{kI} \cdot Index\ Cost + \varepsilon$$

Where the adjusted tracking error to the index is $\sqrt{\sigma_\varepsilon^2}$

Here we subtract only estimated market impact cost (not total estimated cost) for the market adjusted cost since we already adjusted for price appreciation using the stock's underlying beta and index as its proxy.

$$Mkt\ Adj\ Z\text{-}Score = \frac{Pre\text{-}Trade\ Estimate - Mkt\ Adj\ Cost}{Adj\ Tracking\ Error\ to\ the\ Index} \tag{3.24}$$

Adaptation Tactic

Investors also need to evaluate any adaptation decisions employed during trading to determine if traders correctly specify these tactics and to ensure consistency with the investment objectives. For example, many times investors instruct brokers to spread the trades over the course of the day to minimize market impact cost, but if favorable trading opportunities exist then trading should accelerate to take advantage of the opportunity. Additionally, some instructions are to execute over a predefined period of time, such as the next two hours, but with some freedom. In these situations, brokers have the opportunity to finish earlier if favorable conditions exist, or extend the trading period if they believe the better opportunities will occur later in the day.

The main goal of evaluating adaptation tactics is to determine if the adaptation decision (e.g., deviation from initial strategy) was appropriate given the actual market conditions (prices and liquidity). That is, how good a job does the broker do in anticipating intraday trading patterns and favorable trading opportunities.

The easiest way to evaluate adaptation performance is to perform the interval VWAP and interval RPM analyses (see above) over the time period specified by the investor (e.g., a full day or for the specified two hour period) instead of the trading horizon of the trade. This will allow us to determine if the broker actually realized better prices by deviating from the initially prescribed schedule and will help distinguish between skill and luck.

As with all statistical analyses, it is important to have a statistically significant sample size and also perform cross-sectional studies where data points are grouped by size, side, volatility, market capitalization, and market